

Unit 2 Advanced Math Practice Test Module II

1

★ Mark for Review

ABC

For what value of x is the given expression equivalent to $(41k)^{25x}$, where $k > 1$?

$$\sqrt[8]{41k}(\sqrt[9]{41k})^{12}$$

2

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ABC

The expression 4^{18x} is equivalent to k^{3x} , where k is a constant. What is the value of k ?

3

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ABC

The expression $\frac{(6x+8)(3x+6)}{12x+24}$ is equivalent to $\frac{ax+b}{2}$, where a and b are constants and $x > 0$. What is the value of $a+b$?

4

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ABC

Which expression is equivalent to $(3\sqrt{a} - \sqrt{b})^{\frac{2}{7}}$, where $a > b$ and $b > 0$?

A. $\sqrt[7]{9a-b}$

B. $\sqrt[7]{9a-6ab+b}$

C. $\sqrt[7]{9a-3\sqrt{ab}+b}$

D. $\sqrt[7]{9a-6\sqrt{ab}+b}$

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ABC

The expression $12x^2 - mx - 20$, where m is a constant, is equivalent to $(kx - 5)(nx + 4)$, where k and n are integer constants. Which of the following must be an integer?

A. $\frac{12}{m}$

B. $\frac{12}{n}$

C. $\frac{m}{k}$

D. $\frac{m}{n}$

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In the given equation, m and n are constants and $n < 0$. If the equation is true for all values of x , what is the value of $m + n$?

$$4x^2 + mx + 9 = (2x + n)^2$$

A. -15

B. -9

C. -3

D. 12

7

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ABC

The graph of $y = f(x)$ has a y -intercept at $(0, 12)$ and x -intercepts at $(-3, 0)$ and $(2, 0)$. Which of the following equations could define f ?

A. $f(x) = (x + 3)^2(x - 2)$

B. $f(x) = (x + 3)(x - 2)^2$

C. $f(x) = (x - 3)^2(x + 2)$

D. $f(x) = (x - 3)(x + 2)^2$

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ABC

Function f is defined by $f(x) = a\sqrt{x} + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(16, 0)$, and the graph of $y = f(x) - 10$ has an x -intercept at $(9, 0)$. What is the value of b ?

9

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ABC

For the function $f(x) = b\sqrt{a - x} + 7\sqrt{a - x}$, a and b are constants and $f(16) = 0$. If $f(9) > 0$, which of the following could be the value of $a + b$?

A. -3

B. 7

C. 9

D. 15

10

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ABC

What is the sum of the solutions to the given equation?

$$x^2 + 7x + 19 = 28 - 5x$$

A. -12

B. -9

C. -7

D. 12

11

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ABC

What are the solutions to the given equation?

$$4x^2 - 9x - 1 = 0$$

A. $x = \frac{-9 \pm \sqrt{97}}{8}$

B. $x = \frac{-9 \pm \sqrt{65}}{8}$

C. $x = \frac{9 \pm \sqrt{97}}{8}$

D. $x = \frac{9 \pm \sqrt{65}}{8}$

12

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One solution to the given equation can be written as $\frac{\sqrt{k}-1}{2}$, where k is a constant. What is the value of k ?

$$x^2 + x - 5 = 0$$

13

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ABC

In the given equation, k is a positive constant. What is the product of the solutions to the given equation?

$$x^2 - 8kx - (k - 4) = 0$$

A. $4 - k$ B. $k - 4$ C. $-8k$ D. $8k$ **14**

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ABC

If $4x + k$ is a factor of $8x^3 - 26x^2 - 24x$, where k is a positive constant, what is the value of k ?

15

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ABC

In the given equation, p is a positive constant. The solutions to the given equation are integers. Which of the following is (are) a possible value of p ?

$$x^2 + px + 18 = 0$$

- I. 9
- II. 17
- III. 19

A. I only

B. II only

C. I and III only

D. I, II, and III

16

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ABC

The function $f(x) = \frac{1}{2}(x - 5)^2 + 8$ gives the profit, in hundreds of thousands of dollars, a company makes when x thousand units of its product are produced and sold, where $1 \leq x \leq 9$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

A. The company can make a maximum profit of \$500,000.

B. The company can make a maximum profit of \$800,000.

C. The company makes a profit of \$0 when 5,000 units of its product are produced and sold.

D. The company makes a profit of \$0 when 8,000 units of its product are produced and sold.

17

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ABC

The given function $P(x) = -0.03(x - 20)^2 + 12$ models the total profit, in thousands of dollars, that a charity group expects to earn from selling boxes of cookies for x dollars each, where $0 \leq x \leq 40$. Based on the model, which of the following is the best interpretation of $P(10) = 9$ in this context?

A. If each box of cookies is sold for \$9, the total profit will likely be \$10,000.

B. If each box of cookies is sold for \$9, the total profit will likely be \$21,000.

C. If each box of cookies is sold for \$10, the total profit will likely be \$9,000.

D. If each box of cookies is sold for \$10, the total profit will likely be \$22,000.

18

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ABC

In the given system of equations, b is a positive constant. The system has exactly one distinct real solution. What is the value of b ?

$$y = 4x$$

$$y = 3x^2 + bx + 3$$

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ABC

In the given system of equations, p is a constant. The system has exactly one distinct real solution (x, y) . What is the value of y ?

$$y = 2x^2 + 15x + 57$$

$$y = -9x + p$$

20

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ABC

The quadratic function $g(x) = ax^2 + 8x + c$ is defined by the given equation, where a and c are constants. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(0, 0)$ and $(n, 0)$, where n is a nonzero constant. If the maximum value of $g(x)$ is 25, which of the following must be true?

- I. $a < 0$
- II. $c = 0$
- III. $n > 0$

A. I only

B. II only

C. I and II only

D. I, II, and III

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ABC

In the given quadratic function, b and c are constants. The graph of $y = f(x)$ in the xy -plane does not intersect the x -axis. If $f(-7) = f(9)$, which of the following could be the value of $b + c$?

$$f(x) = 3x^2 + bx + c$$

A. -1 B. -3 C. -5 D. -7

22

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ABC

The functions f and g are defined by the given equations, where $x \geq 0$. Which function shows a maximum value as a constant or coefficient for $x \geq 0$?

I $f(x) = 33(0.4)^{x+3}$

II $g(x) = 33(0.16)(0.4)^{x-2}$

A. I only

B. II only

C. I and II

D. Neither I nor II

Answer Key — Unit 2

No.	Answer		No.	Answer
1	7/120		12	21
2	4096		13	A
3	7		14	3
4	D		15	C
5	B		16	B
6	A		17	C
7	B		18	10
8	40		19	39
9	D		20	D
10	A		21	A
11	C		22	B